HYLU User Guide

(Version 20260331)

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HYLU (Hybrid Parallel Sparse LU Factorization) is a general-purpose parallel solver for efficiently solving sparse linear systems ($\mathbf{Ax} = \mathbf{b}$) on multi-core shared-memory architectures. *HYLU* supports solving **unsymmetric**, **symmetric**[1] **indefinite**, and **symmetric positive-definite (SPD)** systems. *HYLU* employs innovative parallel up-looking LU factorization and Cholesky factorization algorithms, which dynamically adapt to varying matrix sparsity patterns by leveraging hybrid numerical kernels. The numerical stability of LU factorization is maintained by combining static pivoting, dynamic diagonal supernode pivoting, dynamic scaling, and iterative refinement.

HYLU delivers high-performance LU factorization and Cholesky factorization for large-scale sparse linear systems from multiple engineering and scientific domains, including circuit simulation, power systems, computational fluid dynamics, electromagnetics, structural analysis, etc. The solver efficiently handles linear systems arising from finite element analysis, 2D/3D modeling, and optimization problems.

HYLU is implemented in `C` and offers seamless integration with `C/C++` applications. The multi-threading of *HYLU* is implemented based on operating system's native threading interface (OpenMP is NOT needed).

1. Functions

HYLU offers 9 groups of functions for solving sparse linear systems. Each group provides both 32-bit and 64-bit integer versions for both real number and complex number systems. The following table lists the format of the *HYLU* function names and their corresponding versions. In the following text, the function names of the 32-bit integer/real number version will be used to explain the functionality of *HYLU*. In the 32-bit integer version, only the input matrix indices use 32-bit integers, while internal LU factor data structures maintain 64-bit integers for scalability. For detailed function prototypes and argument descriptions of the *HYLU* functions, please refer to `<hylu.h>`.

Function name	Version	Library
`HYLU_XXXX`	real number/32-bit integer	libhylu.so|hylu.dll
`HYLU_L_XXXX`	real number/64-bit integer	libhylu_l.so|hylu_l.dll
`HYLU_C_XXXX`	complex number/32-bit integer	libhylu_c.so|hylu_c.dll
`HYLU_CL_XXXX`	complex number/64-bit integer	libhylu_cl.so|hylu_cl.dll

(1) `HYLU_CreateSolver`

This function creates a solver instance, retrieves a pointer to the parameter array, and spawns working threads for parallel execution. Input parameters are initialized to their default values during this process. The number of working threads should not

[1] For *HYLU*, a "symmetric" complex-number matrix means a "Hermitian" matrix.

exceed the number of idle cores.

(2) `HYLU_DestroySolver`

This function frees all allocated memory, terminates the working threads, and destroys the solver instance.

(3) `HYLU_Analyze`

This function performs preprocessing steps including static pivoting (for unsymmetric matrices), matrix reordering, and symbolic factorization, and then builds internal data structures. The input matrix format of *HYLU* is **compressed sparse row (CSR)**, where column indexes in each row are **NOT required to be sorted**. For symmetric matrices, the CSR representation must only store the upper triangular part (including diagonal).

The second argument, '`repeat`', specifies whether linear systems will be solved repeatedly with an identical matrix structure. This is common in applications such as circuit simulation. In such scenarios, **preprocessing is needed only once**. When enabled, this argument impacts the default values of certain parameters. Specifically, in the repeated solving scenario, *HYLU* prioritizes minimizing nonzeros in the LU factors, which may increase preprocessing time but improves subsequent factorization efficiency. In non-repeated solving scenarios, the preprocessing overhead becomes significant, and *HYLU* tends to trade off preprocessing overhead and factorization time.

It is optional to provide the matrix values (the argument '`ax`') to preprocessing, but providing values is strongly recommended for unsymmetric matrices. Without matrix values, static pivoting and static scaling cannot be carried out.

(4) `HYLU_Analyze2`

This function provides similar functionality to `HYLU_Analyze` with user-provided ordering. Static pivoting and static scaling are disabled.

For SPD matrices, the user-provided ordering must guarantee that all diagonal entries of the reordered matrix are non-zero (though this function does not check the numerical positive definiteness). For other matrix types, it is strongly recommended that the user-provided ordering also satisfy this condition, as zero diagonal entries may lead to factorization failure or inaccurate solutions.

(5) `HYLU_Factorize`

This function computes the numerical LU factorization or Cholesky factorization of the preprocessed matrix, decomposing it into a lower triangular matrix ($\mathbf{L}$) and an upper triangular matrix ($\mathbf{U}$ or $\mathbf{L}^{\mathbf{H}}$).

Dynamic diagonal supernode pivoting is performed during LU factorization process. If no suitable pivot can be found in a row or the absolute value of a diagonal element is too small, a pivot perturbation strategy is applied.

(6) `HYLU_Solve`

This function computes the solution vector by performing forward ($\mathbf{Ly} = \mathbf{b}$) and backward ($\mathbf{Ux} = \mathbf{y}$ or $\mathbf{L}^{\mathbf{H}}\mathbf{x} = \mathbf{y}$) substitutions using the computed LU factors.

If pivot perturbation has occurred in the last numerical LU factorization or Cholesky factorization, iterative refinement is automatically performed to correct the solution.

(7) `HYLU_MSolve`

This function solves multiple linear systems that share the same coefficient matrix. If working threads have been created, it performs parallel solves across different right-hand-side vectors.

(8) `HYLU_Determinant`

This function calculates the (approximate) determinant of the matrix. It should be called after numerical factorization has been executed. If pivot perturbation has not occurred in the last numerical factorization, the calculated determinant is exact.

(9) `HYLU_ConditionNumber`

This function calculates the (approximate) 1-norm condition number of the matrix. It should be called after numerical factorization has been executed. The calculated condition number is based on the scaled matrix (if scaling is enabled).

2. Parameters

The following table describes the parameter array. Input parameters are initialized to their default values (marked with an asterisk *) upon calling `HYLU_CreateSolver`. All modifications (if needed) to input parameters should be completed prior to invoking `HYLU_Analyze` or `HYLU_Analyze2`. Refer to Section 5 for the parameter tuning guide.

Parameter	Description	
`parm[0]`: output	Version.	
`parm[1]`: **input**	Timer control. When enabled, `parm[7]` records the wall time of the last function call.	
	0*	Disabled.
	>0	High-precision timer (microsecond precision).
	<0	Low-precision timer (millisecond precision).
`parm[2]`: **input**	Ordering method for fill-in reduction.	
	0*	The ordering method is automatically decided according to the matrix dimension and the '`repeat`' argument of `HYLU_Analyze`.
	1	Approximate minimum degree.
	2	Approximate minimum degree variant.
	3	Nested dissection method 1.
	4	Nested dissection method 2.
	5	Best of 1 and 2.
	6	Best of 3 and 4.
	7	Best of 1-4.

`parm[3]`: **input**	Ordering method switch point. For nested dissection, once a partition size is smaller than `parm[3]`, it switches to a constrained approximate minimum degree method. Smaller `parm[3]` values generally yield better ordering results of nested dissection at the cost of increased ordering time.	
	0*	Automatic control. The value is determined according to the '`repeat`' argument of `HYLU_Analyze`.
	>=64	Allowable range.
`parm[4]`: output	Selected ordering method (1-4), reported by `HYLU_Analyze`.	
`parm[5]`: **input**	Minimum number of columns of a supernode. A supernode must contain at least `parm[5]` columns.	
	32*	Default value.
	>=8	Allowable range.
`parm[6]`: **input**	Maximum number of rows of a supernode. A supernode may contain at most `parm[6]` rows. A larger supernode is split into multiple supernodes.	
	0*	Automatic control. The value is co-determined by whether working threads have been spawned and `parm[23]`.
	>=8	Allowable range.
`parm[7]`: output	Wall time (in **microseconds**) of last function call.	
`parm[8]`: output (only for unsymmetric matrices)	Number of off-diagonal pivots, reported by `HYLU_Factorize`.	
`parm[9]`: output	Number of supernodes, reported by `HYLU_Analyze` or `HYLU_Analyze2`.	
`parm[10]`: **input**	Pivot perturbation coefficient. Zero or small pivots will be replaced by $\text{sign(pivot)} \times 10^{\text{parm[10]}} \times \|\mathbf{A}\|_\infty$.	
	0*	Automatic control (-15 for non-SPD matrices, -10 for SPD matrices, and -12 for symmetric matrices).
	<0	Allowable range.
`parm[11]`: output	Number of perturbed pivots, reported by `HYLU_Factorize`.	
`parm[12]`: output	Current memory usage (in bytes), or required memory size (in bytes) when -4 is returned, reported by `HYLU_Analyze` or `HYLU_Analyze2`.	
`parm[13]`: output	Maximum memory usage (in bytes), reported by `HYLU_Analyze` or `HYLU_Analyze2`.	
`parm[14]`: output	Thread numbers stored in 3 shorts. Use the following method to get them: `const short *threads = (short *)&parm[14];` `threads[0]`: number of physical cores (may be incorrect). `threads[1]`: number of logical cores. `threads[2]`: number of created working threads.	
`parm[15]`: **input**	Maximum number of refinement iterations.	
	0*	Whether to perform iterative refinement and the iteration count are determined automatically.

	>0	If *HYLU* decides to perform iterative refinement, execute `parm[15]` iterations.
	<0	Execute `-parm[15]` iterations.
`parm[16]`: output	Number of refinement iterations performed, reported by `HYLU_Solve` or `HYLU_MSolve`.	
`parm[17]`: output	Number of nonzeros in **L** (including diagonal), reported by `HYLU_Analyze` or `HYLU_Analyze2`.	
`parm[18]`: output (only for unsymmetric matrices)	Number of nonzeros in **U** (excluding diagonal), reported by `HYLU_Analyze` or `HYLU_Analyze2`.	
`parm[19]`: output	Number of floating-point operations of factorization (excluding scaling), reported by `HYLU_Analyze` or `HYLU_Analyze2`.	
`parm[20]`: output	Number of floating-point operations of solving (excluding scaling), reported by `HYLU_Analyze` or `HYLU_Analyze2`.	
`parm[21]`: **input**	Matrix scaling method.	
	>0*	Dynamic matrix value scaling.
	0	Disabled.
	<0	Static matrix value scaling (also dynamic matrix value scaling for SPD matrices).
`parm[22]`: **input** (only for unsymmetric matrices)	Symbolic factorization method. Symmetric symbolic factorization helps reduce preprocessing time, but increases fill-ins for structurally unsymmetric matrices.	
	>0	Unsymmetric symbolic factorization.
	<0	Symmetric symbolic factorization.
	0*	Automatic control.
`parm[23]`: **input**	Consistent symbolic results between sequential and parallel executions.	
	!=0*	Enabled.
	0	Disabled.
`parm[24]`: output	Singular row (a zero row in an unsymmetric matrix or a negative diagonal in an SPD matrix) index, reported when factorization failed with -6.	

3. Return Values

Every function of *HYLU* returns an integer to indicate the error code. This table describes the meanings of the return codes.

Return value	Description
0	Successful.
-1	Invalid instance handle.
-2	Function argument error (e.g., negative matrix dimension, `NULL` pointer).
-3	Invalid input matrix (e.g., matrix indices error).
-4	Out of memory. `parm[12]` will return the needed memory size.

-5	Structurally singular.
-6	Numerically singular or indefinite. `parm[24]` will return the involved row index.
-7	Threads error.
-8	Calling procedure error.
-9	Integer overflow, please use the `HYLU_(C)L_*` functions.
-10	Internal error.

4. Using *HYLU* Library

HYLU is provided as x64 dynamic-link libraries. **Running *HYLU* requires an x64 processor with FMA and AVX2 instruction sets support.** Processors that support the AVX-512 instruction set are recommended.

Please follow the platform-specific instructions below to link the *HYLU* library.

- **Linux**

When linking *HYLU* to the final executable, use the compiler flags `-L<library_path> -l<library_name>` to specify the location and name of the *HYLU* library.

Some system libraries may be required. It is often necessary to add `-lpthread`, `-lm`, and `-ldl` to the linker flags. On older Linux distributions, `-lrt` may also be needed. A typical linking command might look like: `gcc -o myprogram myprogram.c -L/path/to/hylu -lhylu -lpthread -lm -ldl`.

Before running the executable, ensure the system can locate the *HYLU* shared library. The typical method is to set the `LD_LIBRARY_PATH` environment variable: `export LD_LIBRARY_PATH=<library_path>:$LD_LIBRARY_PATH`.

- **Windows (Using Visual Studio)**

Add the directive `#pragma comment(lib, "<library_name>.lib")` anywhere in the source code (commonly near the top). The corresponding `.lib` import library file is needed during the linking stage.

Only the `.dll` file is required to run the executable. The recommended and simplest method is to place the corresponding `.dll` file in the same directory as the executable.

5. Parameter Tuning Guide

The default values of the input parameters are set based on a large number of test cases. Generally speaking, for most practical matrices, accurate solutions can be obtained with high computational performance without adjusting any input parameters. However, in some cases, *HYLU* may get inaccurate solutions or run slow. Please refer to the following guide to tune the parameters.

- **Tuning for Performance**

For some matrices, the ordering method (`parm[2]`) may have a large impact on the fill-ins. If *HYLU* runs slow in factorization, try different ordering methods by adjusting `parm[2]`. The argument '`repeat`' of `HYLU_Analyze` also impacts fill-ins. Setting `repeat = true` and `parm[2] = 0` enables *HYLU* to automatically select the best ordering method from all ordering methods, which tends to minimize the

fill-ins.

Setting `parm[22] > 0` reduces fill-ins for highly-symmetric (but not completely symmetric) matrices, at the cost of increased preprocessing time.

- **Tuning for Accuracy**

Due to the limited pivoting range, for some unsymmetric linear systems, *HYLU* may produce inaccurate solutions. In such cases, consider setting `parm[23] = 0`, adjusting `parm[6]` to a larger value, and utilizing sequential factorization. These configurations expand the pivoting range which helps improve the solution accuracy.

If pivot perturbation has occurred during numerical factorization, `parm[10]` can be adjusted to improve the solution accuracy. `parm[15]` controls the number of refinement iterations. By default, only when pivot perturbation has occurred, iterative refinement will be executed. Setting `parm[15] < 0` will force *HYLU* to execute iterative refinement regardless of whether pivot perturbation has occurred, which may improve the solution accuracy for some linear systems.

Matrix scaling (`parm[21]`) may also affect the solution accuracy.

The current implementation of LDL^H factorization for symmetric indefinite matrices does not perform pivoting. If an accurate solution cannot be found, consider solving the problem as an unsymmetric system (the whole matrix should be provided).

Please note that, if the inaccurate solution is caused by the ill-conditioning of the matrix itself (having a large condition number), then *HYLU* may not solve it accurately regardless of the parameter configuration.

6. Notes

(1) **User must NOT free the memory of the parameter array**, which is managed by *HYLU*.

(2) By default, *HYLU* operates in **row-major order**. For unsymmetric matrices stored in column-major order, set the argument '`mode`' of `HYLU_Solve` to a nonzero value to solve transposed systems ($\mathbf{A}^T\mathbf{x} = \mathbf{b}$) or conjugate transposed systems ($\mathbf{A}^H\mathbf{x} = \mathbf{b}$, only for complex-number systems). However, the parallel scalability of column-wise solve is not as good as that of row-wise solve.

(3) Please ensure that the number of working threads created by *HYLU* is less than or equal to the number of idle cores, otherwise the performance of *HYLU* will degrade severely.

(4) Memory leaks have been observed in *HYLU* libraries linked with Intel MKL BLAS. These leaks are attributed to the MKL BLAS functions themselves.

(5) Setting `parm[23]!=0` ensures consistent symbolic results between sequential and parallel executions, but numerical results may be slightly different. When `parm[23]=0` and multithreading is used, the ordering results from parallel nested dissection are non-deterministic.

(6) The diagonal elements of a complex-number Hermitian matrix must be real numbers.